Domain & Range **TEACHER KEY**

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Domain: all the input values (x) a function is allowed to use.

Range: all the output values (y) the function actually produces.

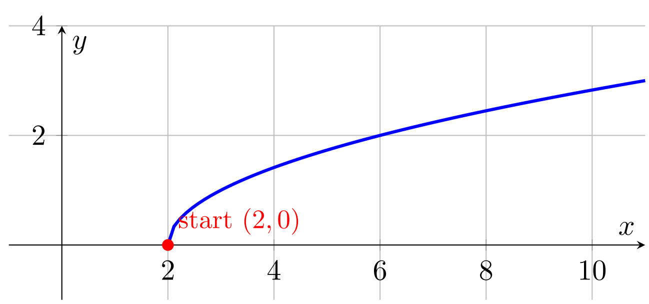
We do together. Find the domain and range of $f(x) = \sqrt{x-2}$.

Step 1. A square root can't hold a negative: set $x - 2 \geq 0$, so the domain is $x \geq 2$.

Step 2. The smallest output is at $x = 2$, where $f(2) = 0$, then it grows.

Step 3. So the range is $y \geq 0$.

See it:



$y = \sqrt{x-2}$ — domain $x \geq 2$, range $y \geq 0$.

You Try. Solve each. Show every step, then name the answer.

a) Give the domain of $f(x) = \dfrac{1}{x-5}$. $x \neq 5$ (all reals except 5)

b) Give the range of $f(x) = x^2 + 1$. $y \geq 1$

Summary (fill in): Watch two traps: a denominator can't be 0, and a square root can't hold a negative.